Sustainability



Sustainability and best practices for water efficiency



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Core questions

- What are best practices for accounting for water use in data centers beyond WUE?

- How can data centers compare water performance across locations and cooling technologies while accounting for effects of climate, energy sources, and appropriate system boundaries?

- With increasing demands for resource efficiency, what does the next generation of innovations look like?



Accounting for Water in Data Center Infrastructure



Cooling Architecture

- Cooling design determines water demand.
- Compare air, evaporative, hybrid, and liquid cooling.



Water Source & Availability

- Evaluate source options and reliability.
- Consider local and seasonal constraints.



Water Stress & Governance

- Assess basin stress and drought risk.
- Understand permits and allocation limits.



Water Quality & Treatment

- Match water quality to system requirements.
- Account for treatment and discharge needs.



Consumption & Metrics

- Distinguish withdrawal from consumption.
- Normalize metrics across system boundaries.



Climate & Reporting

- Climate influences water performance.
- Align metrics with disclosure expectations.



Design implication: Water strategy should be ***source-aware, basin-aware, climate-aware, and technology-aware, not focused solely on water intensity.***

Water accounting and reporting

1. Define the boundary

SITE | OPERATIONS | SCOPE

Operational / site
Cooling, humidification, discharge, domestic, landscaping, testing.

Energy-supply
Grid mix and backup power where water intensity matters.

Construction / commissioning
Initial fill, flushes, dust suppression, and one-time events.

Source context
Potable, reclaimed, brackish, or seawater; note local stress.

Treat small loads separately, but do not hide them.

2. Reconcile the water balance

Meter the incoming and outgoing streams, then reconcile the balance before reporting.

Intake = Evaporation + Drift + Blowdown + Losses – Reuse

Keep commissioning flushes and temporary fills as a separate line item.

Use the same sequence every time: measure, reconcile, normalize.

1. **Meters** — Main, make-up, blowdown, reuse
2. **Reconcile** — Calibration and balance checks
3. **Normalize** — Climate, stress, and energy context

Separate one-time commissioning flushes, temporary fills, and construction water from steady-state operation.

3. Report decision-grade KPIs

Report the KPI set that explains both volume and context.

Metric	What it means	Report as
Withdrawal	What enters the boundary	Annual m3
Consumption	What is not returned locally	Annual m3
Discharge	What leaves the site	Permit class + volume
WUE	Water per IT energy	L/kWh-IT or m3/MWh
Stress-weighted WUE	Basin significance	State the basin factor
Energy-supply water	Upstream electricity burden	Source note

Always include the boundary, denominator, and source context.

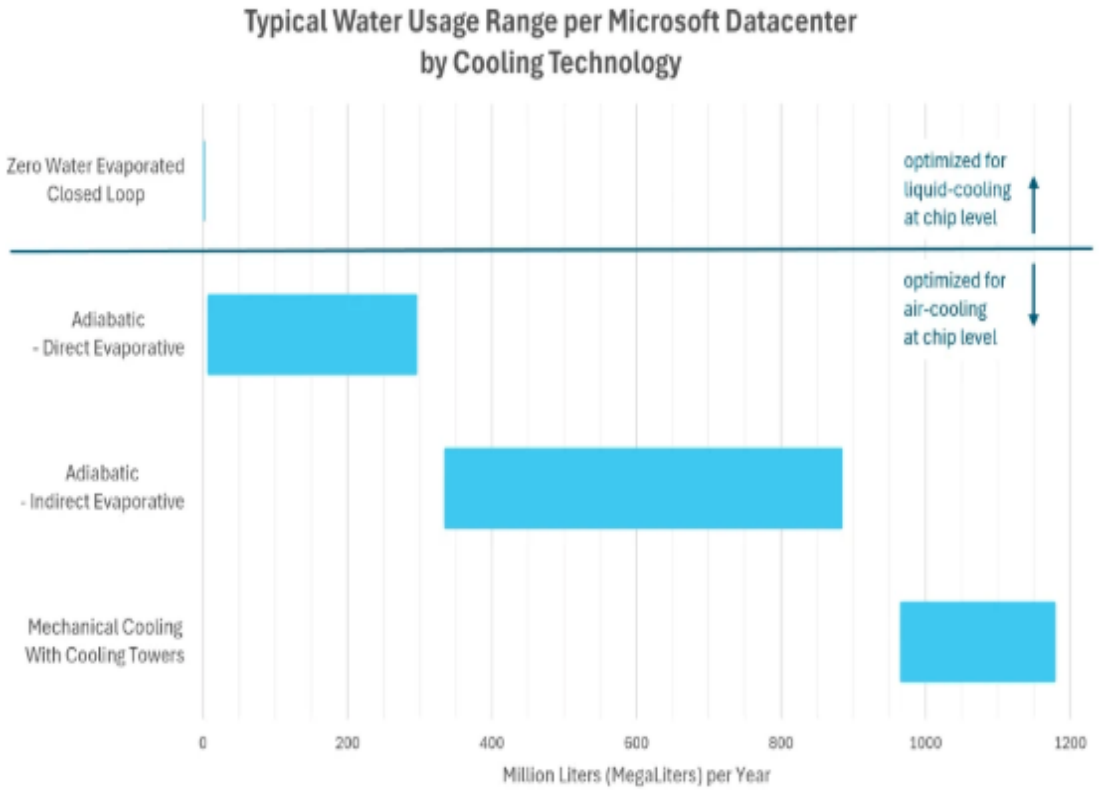
Report in this order

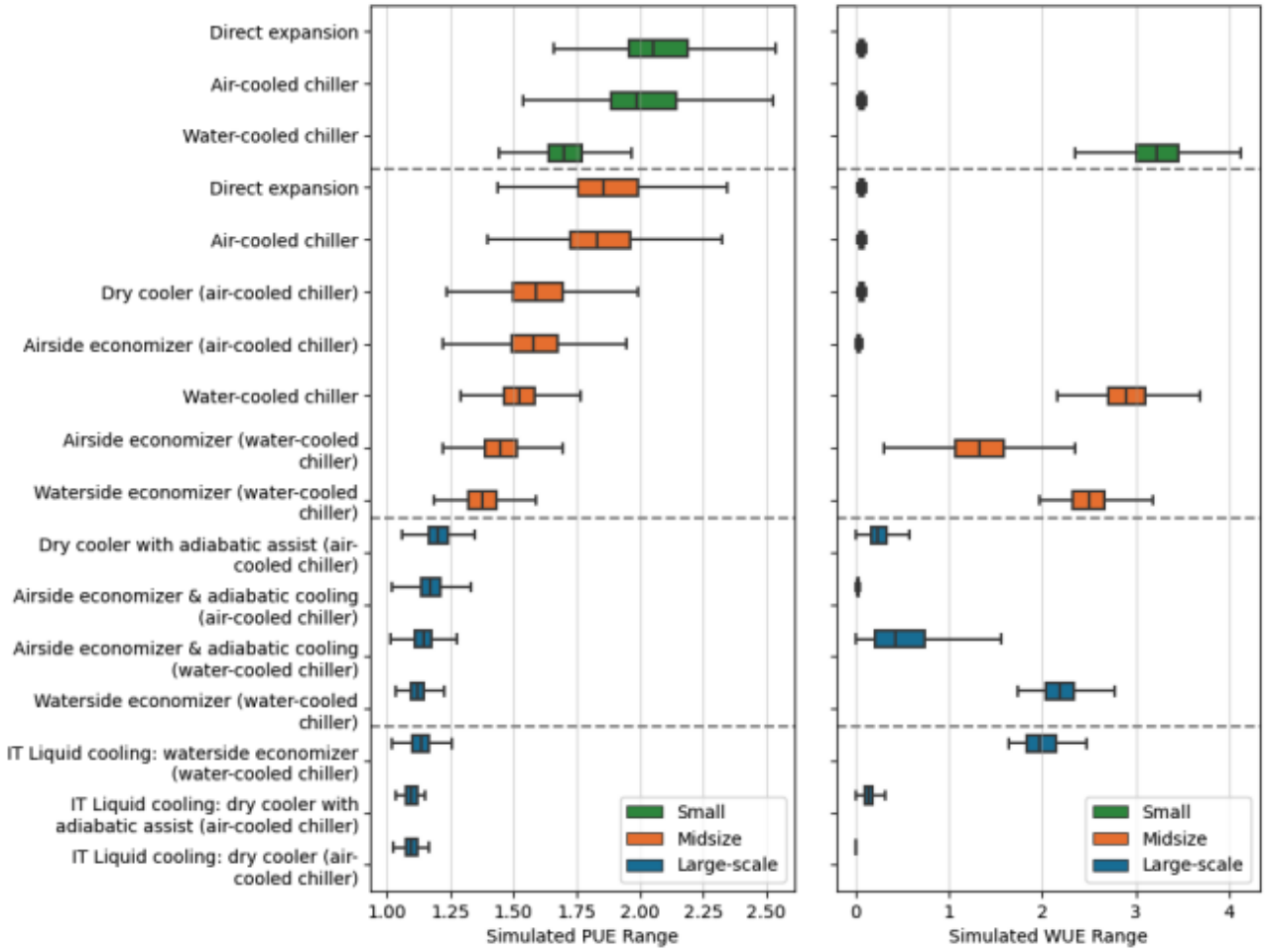
meters → balance → KPI → disclosure

Meters → Balance → KPI → Disclosure

QA: calibration, completeness, uncertainty, permits

Cooling Technology trade offs





Cooling Technology trade offs and best practices

System boundary + water pathway map

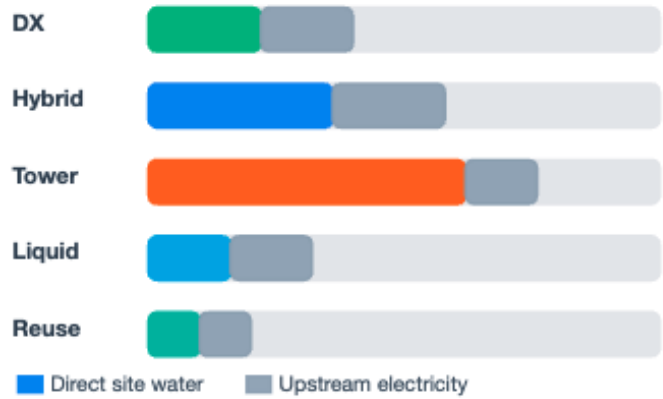
Water sustainability is not just the cooling plant. It spans site loops, treatment, reuse, discharge, and upstream electricity.

Consumptive	Evaporation, blowdown, drift, and losses remove water from the local basin.
Neutral	A practical target: minimize freshwater withdrawal, maximize reuse, and source non-potable water where feasible.
Hidden load	Upstream electricity can carry material embedded water, especially in water-stressed grids.

System	Direct site water	What drives impact	Best practice
DX / air-cooled	**Near-zero direct**	Heat rejected to air; impact shifts upstream to electricity	Use in high-stress basins; meter and leak-check
Hybrid adiabatic	**Low-to-seasonal**	Psychrometric wetting only when needed	Use non-potable water; control by weather + RH
Cooling towers	**Highest direct**	Evaporation + blowdown + drift	Prefer reclaimed water; manage chemistry and CoC
Liquid + dry cooler	**Low direct**	Closed loops; heat rejected with dry air	Strong option for AI density and water restraint
Reclaimed-water loop	**Cuts freshwater draw**	Fit-for-purpose treatment + reuse	Protect against cross-connects and backflow

Illustrative water burden by system type

Higher bars mean more total water burden potential. Actual footprint depends on climate, source quality, CoC, and grid mix.



Recommended best practices

- Define control volumes with a P&ID and separate domestic, process, humidification, discharge, and reuse loops.
- Meter mains, make-up, blowdown, reuse, and any one-time fill/flush volumes; reconcile monthly.
- Use fit-for-purpose treatment: reclaimed, brackish, or non-potable water where source quality permits.
- Tune chemistry to source quality: TDS, hardness, silica, chlorides, and bio-load drive CoC and material selection.
- Add controls and resilience: online sensors, BMS/SCADA, seasonal set-points, dual sources, and emergency storage.

OCP note: "water-neutral" is best treated as a design and operating direction, not a single technology choice.

Cooling technology selection guidance

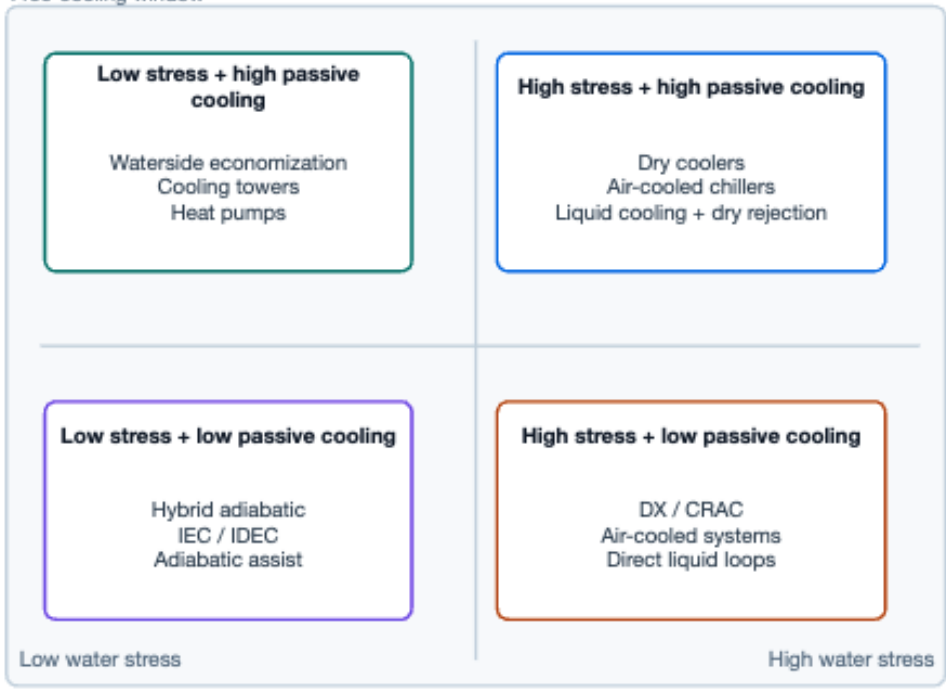
Regional decision table

A practical way to avoid one-size-fits-all cooling choices.

Regional signal	Cooling families to favor	Avoid / limit	Why
High water stress / restrictive discharge	DX, air-cooled chillers, dry coolers, liquid cooling with dry rejection	Cooling towers	Keep direct water use low; use closed loops and dry heat rejection.
Hot climate / modest water stress	Hybrid adiabatic, adiabatic assist, indirect evaporative cooling	Full evaporative towers without non-potable supply	Use water only during peak bins; tune to wet-bulb and source quality.
Cool / humid / low-stress basin	Water-cooled chillers, waterside economization, cooling towers	Over-sizing dry-only plant	Water can buy meaningful energy efficiency where the basin can support it.
Dense AI / heat-reuse market	Direct-to-chip, rear-door HX, immersion + heat pumps	Air-only architectures for very high density	High supply temperatures and closed loops improve reuse and resilience.
Seawater / brackish coastal sites	HX isolation, intake/outfall controls, anti-fouling strategies	Direct open-loop use without permitting review	Source quality, corrosion, plume, and marine safeguards dominate design.

Decision matrix: what drives the choice?

Use region factors to map cooling families instead of picking a default plant.



Key regional factors: basin stress | climate bins | source quality | discharge permits | grid water intensity | heat-reuse market

Guardrails for selection

- Prefer non-potable water where cooling chemistry allows it.
- Pair WUE with WUI and WUEsource to avoid “zero-water” blind spots.
- Size for seasonality, not annual average only.
- Treat heat reuse and reliability as first-class selection criteria.

Call to Action

- Develop best practices and metrics to improve on water sustainability
- Balance water usage and power effectiveness at a localized level
- OCP Water Efficiency Workstream meets biweekly
- Where to find additional information (URL links)

  https://www.opencompute.org/projects/sustainability



Thank You!

